

Reasoning Auditors Beat Fingerprints but Inherit the Negative-Class Problem: A Pilot Benchmark for LLM Paper-Mill Detection

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Artifact: `gunting`, a 9-dimension multi-agent paper-fraud auditor, with a fully reproducible benchmark harness built on the Retraction Watch database and PubMed Central open-access full text.

Abstract

Paper mills and, increasingly, large language model (LLM) text generators produce fluent fraudulent research at a scale that overwhelms peer review. Deployed detectors largely rely on **surface fingerprints** (tortured phrases, template artifacts, metadata anomalies) or single-model classifiers. We ask whether a **decomposed, web-grounded reasoning auditor** improves detection and, critically, whether it can be *evaluated honestly*. We build a reproducible benchmark from the Retraction Watch database (reliable fraud **positives**) and PMC open-access full text, and run a four-cell ablation — decomposition $\times$ web access — plus a lexical fingerprint baseline, on a 14-paper balanced pilot (6 paper-mill retractions, 8 high-integrity-venue controls). Three findings stand out. (1) **Every reasoning configuration separates fluent mills from reputable-venue papers perfectly (AUC = 1.00), while the fingerprint baseline is worse than chance (AUC = 0.44, recall = 0).** (2) The configurations differ not in discrimination but in **calibration**: decomposed + weighted-mean aggregation is markedly more conservative at the strict "likely-fraud" threshold (recall 0.17 vs 0.67), and web grounding does not raise AUC on clean controls but enables catching **fabricated citations** in novel fraud. (3) The hardest and most important problem is the **negative class**: on same-domain *non-retracted* papers, even a no-rubric model flags 5/7 as suspicious — these are either false positives or *undetected mills*, and "non-retracted" is not a safe negative. We additionally show, against freshly **LLM-generated** fabricated papers (the "BadScientist" threat), that the auditor scores fabrications 82–95/100 while clearing an honest synthetic control at 5/100. We release the harness, dataset builder, scorer, and all run artifacts. **Headline numbers (AUC = 1.0) reflect an easy negative class and a small pilot and should not be read as deployment performance**; the contribution is the method, the honest evaluation protocol, and the negative-class diagnosis.

1. Introduction

The scholarly record is under attack from two converging sources. **Paper mills** sell authorship on fabricated manuscripts at industrial scale — Elsevier alone recently exposed 24 bad-actor networks spanning $\sim 1,100$ members, and the 2026 World Conference on Research Integrity named GenAI-accelerated fraud the field's top threat. Simultaneously, **LLM text generation** lets a single actor produce fluent, plausible-but-unsound papers; the *BadScientist* study showed fabricated manuscripts achieving acceptance rates up to 82% against automated reviewers.

Detection has not kept pace. The deployed-detector family — Clear Skies' *Papermill Alarm*, the STM Integrity Hub, Cabanac et al.'s *Problematic Paper Screener* — keys on **surface signals**: tortured phrases ("bosom peril" for breast cancer), template leftovers, citation/metadata anomalies. Recent LLM work (*Pub-Guard-LLM*) frames detection as **retraction classification** with a single model. Both approaches share a blind spot for fluent fraud and, we argue, a deeper **evaluation** problem: they are scored against retraction labels whose *negative* class ("not retracted") is unreliable.

We study `gunting`, an auditor that **decomposes** a paper into nine integrity dimensions (methodology, results, author, duration, findings, summary, formatting, inconsistency, plausibility), audits each with an independent web-enabled agent, and synthesizes a 0–100 fraud-risk score with located, evidence-bearing flags. Our contributions:

1. **A reproducible, leakage-controlled benchmark** built from Retraction Watch + PMC OA full text, with a dataset builder, PMC fetcher, retraction-notice sanitizer, pure-Python scorer (AUC, P/R/F1, ECE, bootstrap CIs), and a lexical fingerprint baseline. One command reproduces every number; the same harness scales to the full ~6,300 fraud-positive corpus.
2. **A clean 2×2 ablation** (decomposition × web) plus baselines, isolating where the auditor's design earns its cost.
3. **The negative-class diagnosis**: a demonstration that benchmark difficulty is governed almost entirely by how negatives are chosen, and that same-domain non-retracted controls are unsafe.
4. **An adversarial arm** against freshly LLM-generated fabrications, testing generalization to fraud with no retraction record and no fingerprint.

We are deliberate about scope: this is a **pilot (n = 14)** with an easy negative class. We report what is real and flag what is not.

2. Related Work

Surface / fingerprint detectors. The Problematic Paper Screener (Cabanac et al.) flags tortured phrases at scale; Clear Skies' Papermill Alarm and the STM Integrity Hub combine lexical and metadata/network signals. These are precise on *crude* mills but, as we show, blind to fluent ones.

LLM detectors. *Pub-Guard-LLM* (2025) detects retracted biomedical articles with three modes (Vanilla Reasoning, RAG, Multi-Agent Debate) over an 11,192-article corpus (24.5% retracted); reported test-set F1 $\approx$ 70, average F1 $\approx$ 76.8 (RAG), best recall $\approx$ 88.7 (Debate), beating fine-tuned SciBERT (F1 $\approx$ 60.9) and few-shot LLMs (F1 $\approx$ 30–48). Our auditor differs in **decomposition** (nine specialist dimensions vs one model) and in producing **located, human-actionable flags**, and we deliberately separate fraud **reasoning** from retraction **status**.

Adversarial generation. *BadScientist* frames the offense — can a generator fool reviewers? We frame the defense and reuse its premise to build an adversarial test set.

3. The `gunting` Auditor

Given one paper's text, `gunting` runs nine **web-enabled specialist agents**, one per integrity dimension, each emitting structured flags (`severity` $\in$ {CRITICAL, HIGH, MEDIUM, LOW}, `claim`, `issue`, `evidence`, `location`, `sources`, `confidence`) and a 0–100 `dimension_risk`. A synthesizer aggregates:

overall = **max(dimension_risk)** if any dimension $\geq$ 80 (a CRITICAL); **else a weighted mean** with integrity dimensions (methodology, results, inconsistency, plausibility) weighted $\times 1.5$ and the rest $\times 1.0$.

Labels: 0–19 *clean*, 20–39 *minor*, 40–69 *serious*, 70–100 *likely fraud*. The aggregation choice matters for calibration (§5.2).

4. Benchmark Construction

Positives (reliable). From the Retraction Watch database (CC0, via Crossref Labs; 9,374 records in our snapshot), we keep `RetractionNature = Retraction` records whose `Reason` contains explicit fraud/mill markers (*Paper Mill*, *Fake/ Compromised Peer Review*, *Fabrication*, *Falsification*, *Unreliable Data*, ...), yielding **6,362 fraud**

positives; 984 carry a PubMed ID, 733 from Hindawi. We draw pilot positives from the high-purity *Paper Mill* / *Fake Peer Review* subset.

The negative-class problem. Retraction Watch certifies *positives*; it does not certify *negatives*. Our first control set — non-retracted PMC papers from the **same domain and era** (2017–2021 oncology miRNA/lncRNA, the genre mills flooded) — proved unsafe: a no-rubric model flagged **5 of 7** as suspicious (§5.4). We therefore fix a **source-quality rule, independent of any score**: controls must come from high-integrity venues (eLife, Nature Communications, Genome Biology, PLoS Biology) with mandatory data-availability and rigorous review. This trades realism (the negatives become *easy*) for label safety, and we report both.

Full text & leakage control. Text is fetched from the PMC BioC API. Because PMC full text for retracted articles can embed a retraction notice — a direct label leak — we **sanitize** every text, stripping lines that announce retraction/withdrawal/ erratum *status*, and drop any paper whose status cannot be cleanly removed. We also instruct every web-enabled agent **never to look up the paper's own retraction status** (Retraction Watch / PubMed) and to judge only from internal evidence + factual verification (do authors/journal/citations exist; are values plausible).

Pilot set. 14 papers: **6** paper-mill retractions + **8** high-integrity-venue controls, domain-adjacent (cancer molecular biology), leak-sanitized.

Configurations (2×2 + baselines).

Config	Decomposition	Web	Note
gunting_web	9 dimensions	yes	full system
gunting_noweb	9 dimensions	no	web ablation
mono_web	single holistic	yes	decomposition ablation
vanilla	single, no rubric	no	LLM floor
fingerprint	—	—	lexical/structural detector

All LLM configs use the same model (Claude Opus 4.x). The fingerprint baseline implements a tortured-phrase lexicon (Problematic-Paper-Screener style) plus placeholder-citation, template, and citation-density heuristics.

5. Results

5.1 Main benchmark (n = 14; 6 positive / 8 negative)

Decision thresholds: **T40** ($\geq$ "serious") and **T70** ($\geq$ "likely fraud"). 95% CIs are 1,000-sample bootstrap.

Config	AUC (CI)	ECE	T40 P / R / F1	T70 P / R / F1
gunting_web	1.00 (1.00–1.00)	0.19	1.00 / 1.00 / 1.00	1.00 / 0.17 / 0.29
gunting_noweb	1.00 (1.00–1.00)	0.17	1.00 / 1.00 / 1.00	1.00 / 0.67 / 0.80
mono_web	1.00 (1.00–1.00)	0.15	1.00 / 1.00 / 1.00	1.00 / 0.67 / 0.80
vanilla	1.00 (1.00–1.00)	0.19	1.00 / 0.83 / 0.91	1.00 / 0.50 / 0.67
fingerprint	0.44 (0.30–0.50)	0.45	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00

Reasoning crushes fingerprinting on fluent mills. Every reasoning config ranks all 6 positives above all 8 controls (positives 52–88; controls 3–17). The lexical detector scores **0 on all 14** papers (AUC 0.44) — the modern oncology mills carry none of the crude tortured-phrase/placeholder signals it depends on. (Sanity check: the same detector scores 50 on a *crude* mill case, PMC9420581, confirming it fires when surface signals exist.)

5.2 Ablations: discrimination vs calibration

On this clean-control set, **decomposition and web access do not change AUC** — all four LLM configs achieve 1.00. The differences are in **operating point**:

- **At T40 (the practical screening threshold), everything works** (F1 0.91–1.00).
- **At T70**, `gunting_web` collapses to recall 0.17, while `gunting_noweb` and `mono_web` hold 0.67. Cause: `gunting`'s **weighted-mean aggregation dilutes** a paper that has many genuinely-clean dimensions (e.g., *duration*, *author*) unless one dimension fires a CRITICAL (≥ 80). True mills cluster in the *serious* band (55–69), correctly suspicious but rarely labeled *likely fraud*.
- **Web access** *lowers* positive scores rather than raising them (e.g., the fabricated-CNN paper PMC9499792: 88 no-web $\rightarrow$ 59 web), because verification confirms the journal/refs are real and the auditor moderates. Web thus trades sensitivity for **specificity and grounding** — its decisive value appears against fabricated citations (§5.5), not on this corpus's AUC.

Implication: the contribution of decomposition is **not** higher AUC on easy negatives; it is (a) **explainability** (located, dimension-attributed flags, §5.3) and (b) **specificity on hard negatives** — and the default fraud threshold and aggregation need recalibration (we suggest threshold 40, or counting dimensions in the *serious*+ band, rather than threshold 70).

5.3 Which dimensions discriminate, and explainability

Mean dimension scores, positives vs controls (`gunting_web`):

Dimension	pos	neg	gap
plausibility	67.2	6.5	60.7
methodology	64.7	4.5	60.2
results	67.2	7.1	60.0
inconsistency	71.5	13.2	58.2
formatting	65.8	8.6	57.2
findings	61.2	5.6	55.5
summary	45.8	3.9	42.0
author	43.0	2.8	40.2
duration	23.7	3.8	19.9

The four **integrity dimensions** weighted $\times 1.5$ in the synthesizer are exactly the most discriminative (gaps ~ 58 –61), empirically validating the design; *duration* contributes least (text-only auditing cannot see submission timelines).

Explainability vs ground truth. Without access to retraction status, `gunting`'s top flags map onto the *documented* Retraction Watch reasons:

Paper	RW reasons (ground truth)	gunting's independent flags
PMC7840786	Paper Mill; Unreliable Data; Concerns about Data	backward miR→FOXO mechanism; no quantitative data; duplicated figure caption
PMC7844839	Paper Mill; Duplication/Plagiarism of Data	template mill family; false ATCC cell-line provenance ; unquantified blots
PMC6910214	Paper Mill; Concerns/Plagiarism of Image	figure-legend mismatches; mechanistically backwards SKP2 claim; impossible "2–3 s" DAB
PMC7656116	Paper Mill; Concerns about Data/Image	cell-line misclassification; cherry-picked TCGA; non-existent siRNA vendor
PMC8072314	Paper Mill; Plagiarism of Data	phantom "miR-203 inhibitor" copy-paste; funding/institution mismatch; impossible 500 μ M
PMC9499792	Paper Mill; Computer-Generated; Referencing	impossible identical 99.6% metrics; off-topic citation padding; tortured CNN terms

The flags are *located* and *checkable*, and recover the integrity failures behind each retraction — the property an editor actually needs.

5.4 The negative-class problem (hard negatives)

Re-running the floor model (`vanilla`) on the discarded **same-domain non-retracted** control set:

- 5 / 7 scored ≥ 40 (serious+); 3 / 7 scored ≥ 70 (likely fraud).

These papers are *not* in Retraction Watch, yet a detector flags most of them. Two readings are possible and **indistinguishable without ground truth we do not have**: they are false positives, *or* they are undetected mills (entirely plausible in this genre and era). Either way, "**non-retracted**" is **not a valid negative label**, and a benchmark's apparent difficulty is set almost entirely by how negatives are chosen. This is, we argue, the central obstacle to honest evaluation of mill detectors — and a reason published retraction-classification scores should be read cautiously.

5.5 Adversarial arm: novel LLM-generated fraud (BadScientist threat)

We generated three papers with an LLM — two **fluent fabrications** (an ML intrusion detector; a biomedical ceRNA study) and one **honest synthetic control** — and audited them **blind** with `gunting_web`. None has a retraction record or surface fingerprint.

Paper	True	gunting	verdict
ADV1 (fabricated ML)	fraud	95	likely_fraud
ADV2 (fabricated biomed)	fraud	82	likely_fraud
ADV_CTRL (honest ML)	clean	5	clean

For ADV1, the auditor **web-verified that the cited journals do not exist** and proved the headline result (frozen-weight transfer across datasets with incompatible feature schemas) is *physically impossible*. For ADV2 it caught mirror-image too-clean fold-changes and the paper's own boast of "uniformly small error bars." The honest control was cleared after verifying its datasets and citations are real. This is the case fingerprints and retraction-lookups cannot address, and where **web-grounded reasoning is decisive**.

6. Comparison to Pub-Guard-LLM (context, not head-to-head)

Pub-Guard-LLM reports test-set $F1 \approx 70$ / avg $F1 \approx 76.8$, recall ≈ 88.7 , on 11,192 articles at 24.5% prevalence, predicting **retraction**. Our reasoning configs reach $F1 = 1.00$ at T40 — but this is **not comparable**: our pilot is

small and balanced, our negatives are *easy* (high-integrity venues), and our target is **fraud reasoning with status-lookup forbidden**, not retraction prediction. The honest takeaway is *not* "gunning beats Pub-Guard"; it is that **scores are dominated by negative-class choice and prevalence**, and that the field needs shared, label-safe negatives before cross-system numbers mean anything.

7. Limitations (read before citing any number)

- **Pilot size.** $n = 14$ (6/8). CIs are wide where they matter (T70 F1 CI for `gunning_web` is [0.0, 0.67]). The full-corpus run (harness provided) is future work.
 - **Easy negatives.** High-integrity-venue controls make $AUC = 1.0$ unsurprising; the honest difficulty lives in §5.4.
 - **Genre skew.** Positives are 5 oncology mills + 1 CS paper; controls skew biomedical. Generalization across fields is untested.
 - **Single model, dual role.** The same model family powers `gunning` and generated the adversarial set — the adversarial result shows Claude can catch Claude-written fraud, not that it generalizes to all generators or that an adversary optimizing against `gunning` could not evade it.
 - **Residual leakage risk.** Two web-enabled agents reported incidentally seeing a retraction mention and stated they ignored it per instructions; we cannot fully guarantee no influence. Sanitization + instruction are mitigations, not proofs.
 - **No image forensics.** Text-only; image duplication (a top RW reason) is inferred indirectly via figure-legend inconsistencies.
 - **Reproducibility caveat.** Exact scores are model/version dependent; the harness reproduces the *protocol* and data, not bit-identical LLM outputs.
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8. Conclusion

On fluent modern paper mills, **reasoning-based auditing decisively outperforms surface fingerprinting** (AUC 1.00 vs 0.44), and a decomposed, web-grounded auditor adds **located, ground-truth-aligned explanations and specificity on hard negatives** — while needing threshold/aggregation recalibration to label fraud rather than merely "serious concern." It also catches **novel LLM-generated fabrications** (82–95/100) that have no retraction record or fingerprint, including by detecting invented citations. But the result that should shape the field is negative: **retraction-based benchmarks cannot certify their negatives**, so headline detection scores — ours included — overstate deployment performance. The next step is a label-safe negative set (adjudicated clean papers, or prospective-then-confirmed cohorts) and a full-corpus run; the harness to do both is released here.

Reproducibility

All code, data builders, run artifacts, and the scorer are in `benchmark/`. See `benchmark/README.md`. The Retraction Watch snapshot is CC0; PMC texts are open-access. `python3 scoring/score.py --labels sets/labeled.jsonl --run ...` regenerates every metric in §5.1.

Artifact and correspondence: `gunning-paper-audit`.